

Self similar diffusion

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1 Navier–Stokes

We begin with the dimensional, incompressible Navier–Stokes equations,

$$\frac{\partial u_i^*}{\partial t^*} + u_j^* \frac{\partial u_i^*}{\partial x_j^*} = -\frac{1}{\rho} \frac{\partial p^*}{\partial x_i^*} + \nu \frac{\partial^2 u_i^*}{\partial x_j^* \partial x_j^*}, \quad (1)$$

together with the continuity equation,

$$\frac{\partial u_i^*}{\partial x_i^*} = 0. \quad (2)$$

We introduce the following nondimensionalization, with velocity and length scales U and L taken to be functions of t :

$$u_i = \frac{u_i^*}{U}, \quad (3)$$

$$x_i = \frac{x_i^*}{L}, \quad (4)$$

$$p = \frac{p^*}{\rho U^2}, \quad (5)$$

$$\frac{dt}{dt^*} = \frac{U^2}{\nu}. \quad (6)$$

We will also need, for each component j ,

$$\frac{dx_j}{dt} = -x_j^* \frac{L'}{L^2} = -x_j \frac{L'}{L}, \quad (7)$$

where prime denotes differentiation with respect to t . Substituting this nondimensionalization and proceeding term by term, the time-derivative term becomes

$$\frac{\partial u_i^*}{\partial t^*} = \left(\frac{dU}{dt} \frac{dt}{dt^*} \right) u_i + U \left(\frac{\partial u_i}{\partial t} \frac{dt}{dt^*} + \frac{\partial u_i}{\partial x_j} \frac{dx_j}{dt} \frac{dt}{dt^*} \right), \quad (8)$$

where the spatial-derivative term is summed over the dummy index j , distinct from the free index i of u_i . Substituting (6) and (7), this becomes

$$\frac{\partial u_i^*}{\partial t^*} = U' \frac{U^2}{\nu} u_i + U \left(\frac{\partial u_i}{\partial t} \frac{U^2}{\nu} - \frac{\partial u_i}{\partial x_j} x_j \frac{L'}{L} \frac{U^2}{\nu} \right). \quad (9)$$

The convection term is

$$u_j^* \frac{\partial u_i^*}{\partial x_j^*} = \frac{U^2}{L} u_j \frac{\partial u_i}{\partial x_j}. \quad (10)$$

The pressure term is

$$\frac{1}{\rho} \frac{\partial p^*}{\partial x_i^*} = \frac{U^2}{L} \frac{\partial p}{\partial x_i}. \quad (11)$$

The viscous term is

$$\nu \frac{\partial^2 u_i^*}{\partial x_j^* \partial x_j^*} = \frac{\nu U}{L^2} \frac{\partial^2 u_i}{\partial x_j \partial x_j}. \quad (12)$$

Substituting (9), (10), (11), and (12) into (1) gives

$$\frac{U^2 U'}{\nu} u_i + \frac{U^3}{\nu} \frac{\partial u_i}{\partial t} - \frac{U^3}{\nu} \frac{L'}{L} x_j \frac{\partial u_i}{\partial x_j} + \frac{U^2}{L} u_j \frac{\partial u_i}{\partial x_j} = -\frac{U^2}{L} \frac{\partial p}{\partial x_i} + \frac{\nu U}{L^2} \frac{\partial^2 u_i}{\partial x_j \partial x_j}. \quad (13)$$

Dividing through by U^3/ν and introducing the instantaneous Reynolds number

$$Re = \frac{UL}{\nu}, \quad \text{so that} \quad \frac{\nu}{UL} = \frac{1}{Re}, \quad (14)$$

gives the nondimensional momentum equation

$$\frac{\partial u_i}{\partial t} = \frac{L'}{L} x_j \frac{\partial u_i}{\partial x_j} - \frac{U'}{U} u_i - \frac{1}{Re} u_j \frac{\partial u_i}{\partial x_j} - \frac{1}{Re} \frac{\partial p}{\partial x_i} + \frac{1}{Re^2} \frac{\partial^2 u_i}{\partial x_j \partial x_j}. \quad (15)$$

Because x_i and u_i are each nondimensionalized by a single, direction-independent factor (L and U , respectively), the continuity equation (2) carries through unchanged,

$$\frac{\partial u_i}{\partial x_i} = 0. \quad (16)$$

Equations (15) and (16) are the nondimensional momentum and continuity equations. The stretching terms $(L'/L) x_j \partial u_i / \partial x_j$ and $-(U'/U) u_i$ have exactly the structure of the terms $\alpha x \partial u / \partial x$ and $-\beta u$ in the model equation (50) derived below for scalar diffusion, which uses the same length and velocity scales L and U ; the remaining terms are the usual advection and pressure terms (each carrying a factor $1/Re$) and the viscous term (carrying $1/Re^2$).

1.1 Fourier-Space Formulation

We take the spatial Fourier transform in the nondimensional coordinate x_i introduced above, with convention

$$\hat{u}_i(k, t) = \int u_i(x, t) e^{-ik \cdot x} dx, \quad u_i(x, t) = \frac{1}{(2\pi)^d} \int \hat{u}_i(k, t) e^{ik \cdot x} dk, \quad (17)$$

where d is the number of spatial dimensions and k is the wavevector conjugate to x . Two standard properties of the transform are used repeatedly below: differentiation becomes multiplication, $\mathcal{F}[\partial u / \partial x_j] = ik_j \hat{u}$, and multiplication by x_j becomes differentiation, $\mathcal{F}[x_j u] = i \partial \hat{u} / \partial k_j$.

Applying (17) to the continuity equation (16) gives

$$k_i \hat{u}_i(k, t) = 0, \quad (18)$$

so that $\hat{u}(k, t)$ is, at every wavevector, orthogonal to k .

We now transform each term of the momentum equation (15) in turn. The time-derivative and decay terms transform directly,

$$\frac{\partial u_i}{\partial t} \longrightarrow \frac{\partial \hat{u}_i}{\partial t}, \quad -\frac{U'}{U} u_i \longrightarrow -\frac{U'}{U} \hat{u}_i. \quad (19)$$

The viscous term transforms using the differentiation rule twice,

$$\frac{1}{Re^2} \frac{\partial^2 u_i}{\partial x_j \partial x_j} \longrightarrow -\frac{k^2}{Re^2} \hat{u}_i, \quad k^2 = k_j k_j. \quad (20)$$

The convective term transforms, using continuity to write $u_j \partial u_i / \partial x_j = \partial(u_i u_j) / \partial x_j$, into a convolution,

$$\frac{1}{Re} u_j \frac{\partial u_i}{\partial x_j} \longrightarrow \frac{ik_j}{Re} \widehat{u_i u_j}(k, t), \quad \widehat{u_i u_j}(k, t) = \frac{1}{(2\pi)^d} \int \hat{u}_i(k', t) \hat{u}_j(k - k', t) dk', \quad (21)$$

and the pressure term transforms into

$$\frac{1}{Re} \frac{\partial p}{\partial x_i} \longrightarrow \frac{ik_i}{Re} \hat{p}(k, t). \quad (22)$$

The stretching term requires both rules together: writing $\partial u_i / \partial x_j \rightarrow ik_j \hat{u}_i$ and then applying the multiplication rule to the result, with j summed as usual,

$$\frac{L'}{L} x_j \frac{\partial u_i}{\partial x_j} \longrightarrow \frac{L'}{L} i \frac{\partial}{\partial k_j} (ik_j \hat{u}_i) = -\frac{L'}{L} \left(d \hat{u}_i + k_j \frac{\partial \hat{u}_i}{\partial k_j} \right) = -\frac{L'}{L} \frac{\partial}{\partial k_j} (k_j \hat{u}_i), \quad (23)$$

which is a dilation operator in wavevector space: the domain stretching in physical space appears in Fourier space as a corresponding rescaling of k .

Collecting (19)–(23), the momentum equation (15) becomes

$$\frac{\partial \hat{u}_i}{\partial t} = -\frac{L'}{L} \frac{\partial}{\partial k_j} (k_j \hat{u}_i) - \frac{U'}{U} \hat{u}_i - \frac{ik_j}{Re} \widehat{u_i u_j}(k, t) - \frac{ik_i}{Re} \hat{p}(k, t) - \frac{k^2}{Re^2} \hat{u}_i, \quad (24)$$

together with the incompressibility constraint (18).

1.2 Leray Projection and the Pressure Equation

Because (18) holds at every wavevector and for all time, its time derivative vanishes as well, $k_i \partial \hat{u}_i / \partial t = 0$. Contracting (24) with k_i therefore isolates a constraint that determines $\hat{p}$, provided the other terms are known once contracted with k_i . Three of them vanish identically. The decay and viscous terms are proportional to $\hat{u}_i$ itself, so $k_i(-U'/U \hat{u}_i) = -(U'/U) k_i \hat{u}_i = 0$ and $k_i(-k^2 \hat{u}_i / Re^2) = 0$ by (18). The stretching term also vanishes on contraction: differentiating the identity $k_i \hat{u}_i(k, t) = 0$ with respect to k_j gives

$$\hat{u}_j + k_i \frac{\partial \hat{u}_i}{\partial k_j} = 0, \quad (25)$$

and using (18) and (25) together shows

$$k_i \frac{\partial}{\partial k_j} (k_j \hat{u}_i) = 0, \quad (26)$$

so the stretching term, like the decay and viscous terms, is already orthogonal to k and contributes nothing to the pressure. This is the Fourier-space counterpart of the fact, checked directly in physical space, that the divergence of each of these terms vanishes identically once (16) holds: the self-similar stretching and decay of the frame are purely kinematic and do not disturb the incompressibility constraint.

Only the convective and pressure terms survive the contraction, giving

$$0 = -\frac{ik_i k_j}{Re} \widehat{u_i u_j}(k, t) - \frac{ik^2}{Re} \hat{p}(k, t), \quad (27)$$

so that, for $k \neq 0$,

$$\hat{p}(k, t) = -\frac{k_i k_j}{k^2} \widehat{u_i u_j}(k, t). \quad (28)$$

This is the Fourier-space pressure Poisson equation: the Reynolds number cancels out of it, and the self-similar terms drop out entirely, so $\hat{p}$ retains exactly the classical incompressible form, evaluated on the instantaneous velocity field in the self-similar frame. (The $k = 0$ mode is undetermined by (27), but also does not appear in (24) at $k = 0$, so no information is lost.)

1.3 The Pressure-Free Momentum Equation

Substituting (28) into the pressure term of (24) and combining it with the convective term gives

$$-\frac{ik_j}{Re} \widehat{u_i u_j} - \frac{ik_i}{Re} \hat{p} = -\frac{i}{Re} \left(\delta_{ij} - \frac{k_i k_j}{k^2} \right) k_l \widehat{u_j u_l}(k, t) = -\frac{i}{Re} P_{ij}(k) k_l \widehat{u_j u_l}(k, t), \quad (29)$$

where

$$P_{ij}(k) = \delta_{ij} - \frac{k_i k_j}{k^2} \quad (30)$$

is the Leray projector onto the subspace orthogonal to k ; it is symmetric, idempotent ($P_{ij}P_{jl} = P_{il}$), and satisfies $k_i P_{ij} = 0$, consistent with (18). Substituting (29) into (24) eliminates the pressure entirely and leaves a single closed equation for $\hat{u}_i(k, t)$,

$$\frac{\partial \hat{u}_i}{\partial t} = -\frac{L'}{L} \frac{\partial}{\partial k_j} (k_j \hat{u}_i) - \frac{U'}{U} \hat{u}_i - \frac{i}{Re} P_{ij}(k) k_l \widehat{u_j u_l}(k, t) - \frac{k^2}{Re^2} \hat{u}_i, \quad (31)$$

subject to $k_i \hat{u}_i(k, t) = 0$, which (31) preserves automatically since every term on the right is orthogonal to k . Equation (31) is the Fourier-space, pressure-free form of the nondimensional, self-similar Navier–Stokes equations: it is identical to the standard projected spectral Navier–Stokes equation, with the classical viscous term $-k^2 \hat{u}_i / Re^2$ and Leray-projected convective term, together with the two additional terms $-\frac{L'}{L} \frac{\partial}{\partial k_j} (k_j \hat{u}_i)$ and $-\frac{U'}{U} \hat{u}_i$ that encode, respectively, the rescaling of the wavevector and the decay of the velocity scale associated with the self-similar frame.

1.4 The Stretching and Decay Terms Alone

Consider the two terms of (31) that arise from the self-similar frame itself, rather than from the Navier–Stokes physics, isolated as a standalone linear PDE for $\hat{u}_i(k, t)$,

$$\frac{\partial \hat{u}_i}{\partial t} = -\frac{L'}{L} \frac{\partial}{\partial k_j} (k_j \hat{u}_i) - \frac{U'}{U} \hat{u}_i. \quad (32)$$

Unlike the convective and pressure terms, the coefficients L'/L and U'/U depend only on t , not on k or x , so no convolution or projection is needed to move between representations: (32) is exactly the Fourier transform, term by term, of the two terms of (15) that produced it in the first place,

$$\frac{\partial u_i}{\partial t} = \frac{L'}{L} x_j \frac{\partial u_i}{\partial x_j} - \frac{U'}{U} u_i. \quad (33)$$

So (32) does transform cleanly back to physical space, with none of the nonlocality that pressure elimination required.

It is worth confirming this by solving each equation independently with the method of characteristics and checking that the resulting solutions are Fourier transforms of one another. In physical space, (33) is carried along the same characteristic already found in (7), $dx_j/dt = -x_j L'/L$, along which $du_i/dt = -(U'/U) u_i$. Writing $\mu = \mu(t; t_0) \equiv L(t)/L(t_0)$, the characteristic gives $x_j(t_0) = \mu x_j(t)$ and the amplitude equation integrates to $u_i(t) = [U(t_0)/U(t)] u_i(t_0)$, so that

$$u_i(x, t) = \frac{U(t_0)}{U(t)} u_i(\mu x, t_0). \quad (34)$$

In wavevector space, (32) expands by the product rule, as in (23), to

$$\frac{\partial \hat{u}_i}{\partial t} + \frac{L'}{L} k_j \frac{\partial \hat{u}_i}{\partial k_j} = -\left(d \frac{L'}{L} + \frac{U'}{U}\right) \hat{u}_i. \quad (35)$$

Its characteristics, $dk_j/dt = (L'/L) k_j$, scale by μ in k -space, $k_j(t_0) = k_j(t)/\mu$, the reciprocal of the $1/\mu$ scaling found in physical space – exactly the inverse relationship that Fourier duality imposes between conjugate variables. Integrating the amplitude equation along this characteristic gives

$$\hat{u}_i(k, t) = \mu^{-d} \frac{U(t_0)}{U(t)} \hat{u}_i\left(\frac{k}{\mu}, t_0\right). \quad (36)$$

Fourier transforming (34) directly, using $\mathcal{F}[f(\mu x)](k) = \mu^{-d} \hat{f}(k/\mu)$ for $\mu > 0$, reproduces (36) exactly. The two solutions agree term for term, confirming that (32) transforms cleanly back to (33). This isolated sub-equation is the vector counterpart, for the velocity field, of the pair of scalar sub-problems (??) and (??) solved by characteristics further below for the model diffusion equation; here the stretching and decay act jointly on each velocity component, rather than being split into separate fractional steps.

2 1D Diffusion equation

2.1 Governing equations

The 1D diffusion equation is given by

$$\frac{\partial u^*}{\partial t^*} = D \frac{\partial^2 u^*}{\partial (x^*)^2}. \quad (37)$$

We use the following non-dimensionalization

$$u = \frac{u^*}{U}, \quad (38)$$

$$x = \frac{x^*}{L_k}, \quad (39)$$

where U and L_k are functions of time. For the temporal coordinate, we define

$$t^* = s(t), \quad (40)$$

such that

$$\frac{ds}{dt} = s' = \frac{L_k^2}{D}. \quad (41)$$

Because

$$\frac{dt^*}{dt^*} = 1 = \frac{ds}{dt} \frac{dt}{dt^*} \quad (42)$$

it follows that

$$\frac{dt}{dt^*} = \frac{D}{L_k^2}. \quad (43)$$

Using the chain rule, the left-hand side of (37) becomes

$$\frac{\partial(uU)}{\partial t^*} = \frac{\partial(uU)}{\partial t} \frac{dt}{dt^*} + U \frac{\partial u}{\partial x} \frac{\partial x}{\partial t} \frac{dt}{dt^*}. \quad (44)$$

Differentiating (39) with respect to t at fixed x^* yields

$$\frac{dx}{dt} = -\frac{x^* L'_k}{L_k^2} = -\frac{L'_k}{L_k} x. \quad (45)$$

Substituting (45) and (43) into (44) gives

$$\begin{aligned} \frac{\partial(uU)}{\partial t^*} &= \left(U \frac{\partial u}{\partial t} + uU' - U \frac{L'_k}{L_k} x \frac{\partial u}{\partial x} \right) \frac{D}{L_k^2} \\ &= U \left(\frac{\partial u}{\partial t} + u \frac{U'}{U} - \frac{L'_k}{L_k} x \frac{\partial u}{\partial x} \right) \frac{D}{L_k^2}. \end{aligned} \quad (46)$$

We now transform the right-hand side of (37):

$$D \frac{\partial^2(uU)}{\partial (x^*)^2} = DU \frac{\partial^2 u}{\partial (x^*)^2} = DU \frac{\partial}{\partial x^*} \left(\frac{\partial u}{\partial x} \frac{\partial x}{\partial x^*} \right). \quad (47)$$

Substituting the spatial derivative scale

$$\frac{dx}{dx^*} = \frac{1}{L_k}, \quad (48)$$

and simplifying, we obtain

$$D \frac{\partial^2(uU)}{\partial(x^*)^2} = DU \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \frac{1}{L_k} \right) \frac{1}{L_k} = \frac{DU}{L_k^2} \frac{\partial^2 u}{\partial x^2}. \quad (49)$$

Equating (46) and (49) and dividing by common factors, we arrive at the transformed equation:

$$\frac{\partial u}{\partial t} = \frac{L'_k}{L_k} x \frac{\partial u}{\partial x} - \frac{U'}{U} u + \frac{\partial^2 u}{\partial x^2}. \quad (50)$$

We assume that U is of the form

$$U = (t^* + \alpha_U)^{\beta_U}, \quad L = (t^* + \alpha_L)^{\beta_L}. \quad (51)$$

It follows that

$$\frac{U_{k+i}}{U_k} = \left(1 + \frac{i \cdot \Delta t_k^*}{t_k^* + \alpha_U} \right)^{\beta_U}, \quad \frac{L_{k+i}}{L_k} = \left(1 + \frac{i \cdot \Delta t_k^*}{t_k^* + \alpha_L} \right)^{\beta_L}. \quad (52)$$

We can express Δt^* using

$$\Delta t_k^* = \int_{t_k}^{t_{k+1}} \frac{L^2}{D} dt \approx \frac{\Delta t}{12D} (23L_k^2 - 16L_{k-1}^2 + 5L_{k-2}^2). \quad (53)$$

As such

$$\frac{U_{k+i}}{U_k} = \left(1 + \frac{i \cdot \Delta t}{12D} \frac{23L_k^2 - 16L_{k-1}^2 + 5L_{k-2}^2}{U_k^{1/\beta_U}} \right)^{\beta_U}, \quad \frac{L_{k+i}}{L_k} = \left(1 + \frac{i \cdot \Delta t}{12D} \frac{23L_k^2 - 16L_{k-1}^2 + 5L_{k-2}^2}{L_k^{1/\beta_L}} \right)^{\beta_L}. \quad (54)$$

We store $\ln(U_k)$ and $\ln(L_k)$ instead of U_k and L_k to avoid overflow, but naively substituting $U_k^{1/\beta_U} = \exp[\ln(U_k)/\beta_U]$ and $L_k^2 = \exp[2\ln(L_k)]$ into (54) defeats the purpose: exponentiating $2\ln(L_k)$ or $-\ln(U_k)/\beta_U$ just hands back the huge quantity we were trying to avoid. Instead we combine every sum and difference in log space using

$$\log_b(X + Y) = \log_b(X) + \log_b(1 + b^{y-x}), \quad \log_b(X - Y) = \log_b(X) + \log_b(1 - b^{y-x}) \quad (X > Y > 0), \quad (55)$$

where $x = \log_b(X)$, $y = \log_b(Y)$.

2.2 Integration

Let

$$\Phi_{\Delta t}^{(D)}(u_k) = u_k + \int_{t_k}^{t_k + \Delta t} \frac{\partial^2 u}{\partial x^2} dt, \quad (56)$$

and

$$\Phi_{\Delta t}^{(S)}(u_k) = u_k + \int_{t_k}^{t_{k+1}} \left(\frac{L'}{L} x \frac{\partial u}{\partial x} - \frac{U'}{U} u \right) dt. \quad (57)$$

2.2.1 scaling operator

evaluating $\Phi_{\Delta t}^{(S)}$ can be done by solving the following pde

$$\frac{\partial u}{\partial t} = x \frac{L'}{L} \frac{\partial u}{\partial x} - \frac{U'}{U} u, \quad (58)$$

with the method of characteristics. Along a curve $x(t)$ satisfying

$$\frac{dx}{dt} = -x \frac{L'}{L}, \quad (59)$$

the total derivative of u is

$$\frac{du}{dt} = \frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} \frac{dx}{dt} = \frac{\partial u}{\partial t} - x \frac{L'}{L} \frac{\partial u}{\partial x}, \quad (60)$$

so along this characteristic the PDE reduces to the ODE

$$\frac{du}{dt} = -\frac{U'}{U}u. \quad (61)$$

Integrating (59) from t_k to t_{k+1} gives

$$\ln x_{k+1} - \ln x_k = -[\ln L_{k+1} - \ln L_k] \implies x_{k+1} = x_k \frac{L_k}{L_{k+1}}, \quad (62)$$

so that the characteristic arriving at x_{k+1} originated at

$$x_k = x_{k+1} \frac{L_{k+1}}{L_k}. \quad (63)$$

Integrating (61) in the same way gives

$$u_{k+1} = u_k \frac{U_k}{U_{k+1}}. \quad (64)$$

Combining this with the initial condition $u_k(x)$ and the characteristic origin (63) gives the solution of the PDE at $t_{k+1} = t_k + \Delta t$, i.e. the scaling operator in closed form,

$$\Phi_{\Delta t}^{(S)}(u_k)(x) = u_{k+1}(x) = \frac{U_k}{U_{k+1}} u_k\left(x \frac{L_{k+1}}{L_k}\right). \quad (65)$$

Substituting in

$$\tilde{U}_{k+1} = \frac{U_{k+1}}{U_k}, \quad (66)$$

$$\tilde{L}_{k+1} = \frac{L_{k+1}}{L_k}, \quad (67)$$

we have

$$\Phi_{\Delta t}^{(S)}(u_k)(x) = u_{k+1}(x) = \frac{1}{\tilde{U}_{k+1}} u_k\left(x \tilde{L}_{k+1}\right). \quad (68)$$

$\tilde{U}_{k+1}$ and $\tilde{L}_{k+1}$ are set such that the resulting solution has a specific first order moment, M_0 and second order moment, M_2 . Denote the moments of the input u_k as

$$M_0^* = \int u_k dx, \quad (69)$$

and

$$M_2^* = \int x^2 u_k dx. \quad (70)$$

The requirement for the first moment

$$M_0 = \int \frac{1}{\tilde{U}_{k+1}} u_k\left(x \tilde{L}_{k+1}\right) dx = \frac{1}{\tilde{U}_{k+1} \tilde{L}_{k+1}} M_0^*. \quad (71)$$

The second moment requirement is

$$M_2 = \frac{1}{M_0} \int x^2 \frac{1}{\tilde{U}_{k+1}} u_k\left(x \tilde{L}_{k+1}\right) dx. \quad (72)$$

Substituting $y = x \tilde{L}_{k+1}$ in the integral gives

$$\int x^2 \frac{1}{\tilde{U}_{k+1}} u_k\left(x \tilde{L}_{k+1}\right) dx = \frac{1}{\tilde{U}_{k+1} \tilde{L}_{k+1}^3} \int y^2 u_k(y) dy = \frac{M_2^*}{\tilde{U}_{k+1} \tilde{L}_{k+1}^3}, \quad (73)$$

so that (72) becomes

$$M_2 = \frac{M_2^*}{M_0 \tilde{U}_{k+1} \tilde{L}_{k+1}^3}. \quad (74)$$

Using the first-moment requirement in the form $M_0 \tilde{U}_{k+1} \tilde{L}_{k+1} = M_0^*$ to eliminate $\tilde{U}_{k+1}$, this reduces to

$$M_2 = \frac{M_2^*}{M_0^* \tilde{L}_{k+1}^2}, \quad (75)$$

which gives

$$\tilde{L}_{k+1} = \sqrt{\frac{M_2^*}{M_0^* M_2}}. \quad (76)$$

Substituting (76) back into the first-moment requirement $\tilde{U}_{k+1} \tilde{L}_{k+1} = M_0^*/M_0$ then yields

$$\tilde{U}_{k+1} = \frac{M_0^*}{M_0 \tilde{L}_{k+1}} = \frac{(M_0^*)^{3/2}}{M_0} \sqrt{\frac{M_2}{M_2^*}}. \quad (77)$$

3 Post-processing

Assume that L and U are of the form

$$L = (t^* + a_L)^{b_L}, \quad (78)$$

and

$$U = (t^* + a_U)^{b_U}. \quad (79)$$

Define

$$\mathcal{L}_k = \ln L_k, \quad \mathcal{U}_k = \ln U_k, \quad (80)$$

together with the per-step log-ratios

$$\ell_k = \ln \tilde{L}_{k+1}, \quad \lambda_k = \ln \tilde{U}_{k+1}. \quad (81)$$

$$\mathcal{L}_k = \mathcal{L}_0 + \sum_{i=1}^k \ell_i = \ln(L_k), \quad \mathcal{U}_k = \mathcal{U}_0 + \sum_{i=1}^k \lambda_i = \ln(U_k). \quad (82)$$

grows only linearly in k ; unlike L_k and U_k themselves, $\mathcal{L}_k$ and $\mathcal{U}_k$ never overflow.

Differentiating (78) and (79) with respect to t^* gives

$$t^* + a_L = L^{1/b_L}, \quad t^* + a_U = U^{1/b_U}. \quad (83)$$

By the chain rule

$$\alpha = \frac{\frac{dL}{dt^*} \frac{dt^*}{dt}}{L} = \frac{b_L}{t^* + a_L} \frac{L^2}{D}. \quad (84)$$

Substituting (83)

$$\alpha = \frac{b_L}{D} L^{2-1/b_L}. \quad (85)$$

The same steps applied to β give

$$\beta = \frac{b_U}{D} L^2 U^{-1/b_U}. \quad (86)$$

Taking logarithms

$$\ln \alpha_k = \ln \frac{b_L}{D} + \left(2 - \frac{1}{b_L}\right) \mathcal{L}_k, \quad (87)$$

$$\ln \beta_k = \ln \frac{b_U}{D} + 2\mathcal{L}_k - \frac{\mathcal{U}_k}{b_U}. \quad (88)$$

Finally, α_k and β_k are estimated directly from the log-ratios already in hand, without any finite difference of L_k or U_k themselves,

$$\alpha_k \approx \frac{\ell_k}{\Delta t}, \quad \beta_k \approx \frac{\lambda_k}{\Delta t}. \quad (89)$$

By fitting curves to Eq. (87) and Eq. (88) we can compute a_L , a_U , b_L , and b_U .